Syllabus

ENPM663 AI-Augmented Robot Software Systems

Spring 2027

Section:	0101 & RO01	**Professor:**	Z. Kootbally
Credits:	3	**Pronouns:**	He/Him
Course Dates:	TODO: verify Spring 2027 dates	**Email:**	zeidk@umd.edu
Course Times:	TODO: verify weekly meeting time	**Office Hours:**	By appointment
Classroom:	TODO: assigned room		

Course Description

This hands-on course examines how modern AI techniques, including large language models (LLMs), vision-language models (VLMs), and learned manipulation components, are integrated into ROS 2 based robotic systems. Students study classical task and motion planning, then extend each layer with AI-augmented counterparts: LLM-based task planning, VLM grounding for scene understanding, and learned grasp prediction for manipulation. The course is structured around the Agile Robotics for Industrial Automation Competition (ARIAC) 2026 (https://pages.nist.gov/ARIAC_docs/en/latest/) as a unifying testbed, but the concepts and engineering patterns taught are general to autonomous robotic systems.

Students entering this course are expected to have working knowledge of ROS 2 (publishers and subscribers, services, actions, parameters, launch files). Week 1 provides a refresher on the ROS 2 topics most directly used in the course (lifecycle nodes, TF2, QoS, executors). A significant amount of Python programming is required, with selected use of C++ for components where it is the industry standard.

Learning Outcomes

After successfully completing this course you will be able to:

- Design system architectures that integrate classical robotic components (perception, planning, control) with AI-based components (LLMs, VLMs, learned models).
- Implement behavior trees in `py_trees_ros` for robust task execution, including failure handling and recovery.
- Apply classical task-level planning using PDDL to generate executable plans for robotic systems.
- Build LLM-based task planners that produce structured plans via prompting, structured output, and tool/function calling.
- Use vision-language models to ground robotic plans in observed scenes and recover from perception failures.
- Integrate pre-trained learned grasp prediction models into a ROS 2 manipulation pipeline.

- Plan and execute robot motion using MoveIt2 for pick-and-place and assembly tasks.
- Evaluate AI-augmented robotic systems using quantitative metrics for plan validity, grounding accuracy, grasp success, and end-to-end task performance.

Required Resources

- **Course Website:** `elms.umd.edu`
- **Operating system:** Ubuntu 24.04.x LTS (Noble Numbat): https://releases.ubuntu.com/noble/
- **Middleware:** ROS 2 Jazzy Jalisco: https://docs.ros.org/en/jazzy/Installation.html
- **Simulator:** Gazebo Harmonic: https://gazebosim.org/docs/harmonic/
- **Editor:** Visual Studio Code: https://code.visualstudio.com/
- **ARIAC 2026:** https://pages.nist.gov/ARIAC_docs/en/latest/
- **AI access:** Either a paid LLM API account (e.g., Anthropic Claude, OpenAI, Google Gemini) or local open-weight models (e.g., Llama 3.1, Qwen 2.5). Course assignments will accept either; cost guidance will be provided per assignment.

Prerequisites

Working knowledge of ROS 2 is required. Specifically, students should be comfortable with:

- Creating and managing ROS 2 packages and workspaces.
- Writing publishers, subscribers, services, and action clients and servers in Python.
- Using launch files, parameters, and basic command-line tools (`ros2 topic`, `ros2 node`, etc.).
- Reading C++ ROS 2 code (writing C++ is not a prerequisite, but reading it is expected).

A diagnostic quiz in Week 1 helps students and the instructor calibrate on ROS 2 background. The quiz is ungraded.

Course Structure

- Before class meetings, students need to consolidate the previous lessons since each week builds directly on the previous one. Early in the course, expect roughly 4 hours per week on assignments. As the course progresses, assignments and the final project may require 6 to 8 hours per week.
- Quizzes are administered in class (unless you are registered as a remote student). Absent students cannot take quizzes.
- Class meetings combine lecture and in-class exercises. Bring a laptop to every meeting. Attendance and active participation are not graded but are strongly encouraged.
- Late submissions incur a grading penalty. Extensions are granted only when more than one group needs additional time on a submission.

- No quizzes or assignments will be dropped. The final project is the only required individual deliverable at the end of the semester; there is no alternative to the final project grade.
- The final project is evaluated on an unseen ARIAC scenario. Students run their integrated system in class, and the performance is graded.
- A final project presentation is required at the end of the semester. Students discuss their approach, design decisions, and results.
- Group projects use a peer review system. Each student in a group rates the contributions of other group members.

Tips for Success in this Course

1. **Participate.** Engage with the lectures, ask questions, and discuss course content with your classmates. Robotic systems are integration problems; you will learn faster by talking through design decisions with peers.
2. **Manage your time.** The course moves quickly. Block out time in your schedule for assignments and include extra time for environment and dependency issues.
3. **Log in regularly.** Check ELMS-Canvas several times a week for announcements and discussion replies. Check daily near group submission deadlines.
4. **Do not fall behind.** Each week builds on the previous one. If you feel you are falling behind, contact the instructor early.
5. **Use ELMS notification settings.** Enable instant or daily notifications so you do not miss announcements.
6. **Ask for help.** For ELMS or technology issues, contact IT support. For course content, contact the instructor or classmates.

Policies and Resources for Graduate Courses

It is our shared responsibility to know and abide by the University of Maryland policies that apply to all courses, including:

- Academic integrity
- Student and instructor conduct
- Accessibility and accommodations
- Attendance and excused absences
- Grades and appeals
- Copyright and intellectual property

See `https://gradschool.umd.edu/course-related-policies` for the full set of graduate course-related policies.

Names, Pronouns, and Self-Identifications

The University of Maryland recognizes the importance of a diverse student body and is committed to fostering inclusive and equitable classroom environments. You are invited, if you wish, to tell the instructor how you want to be referred to in this class, both in terms of your name and pronouns. The pronouns someone uses are not necessarily indicative of their gender identity. Visit `lgbtq.umd.edu` to learn more.

Whether and how to disclose how you identify in terms of gender, race, class, sexuality, religion, ability, and other aspects of your identity is your choice and should be self-identified, not presumed or imposed.

Communication with Instructor

- **Email.** Please email `zeidk@umd.edu` for personal, academic, or intellectual concerns. Please do not email about questions that are easily answered by the syllabus or ELMS (e.g., due dates, point values). Responses are typically sent within 24 hours during weekdays.
- **ELMS.** Important announcements (including assignment changes and due-date changes) are posted on ELMS. Enable email and announcement notifications to avoid missing messages.
- **Communication with peers.** With a diversity of perspectives and experience, disagreement and debate are expected. Conduct yourself professionally and respectfully. This course is not a space for hate speech, harassment, or derogatory language. Alert the instructor immediately if you feel threatened, dismissed, or silenced.

Major Assignments

- **Individual Assignments.** Individual assignments reinforce and apply the concepts covered in class independently. Detailed instructions and submission guidelines are provided for each assignment.
- **Quizzes.** Quizzes assess your understanding of course material. They are administered at the beginning of class, either on Canvas or on paper, and typically last 10 to 20 minutes. All quizzes are closed-notes.
- **Participation and Engagement.** Active participation enhances your understanding and creates a dynamic learning environment.
- **Team Projects.** Team projects apply course concepts to integration problems. Each team member is expected to contribute actively, ensuring equitable participation and shared responsibility.
- **Final Project.** The final project integrates the components covered throughout the course (behavior trees, classical planning, LLM-based planning, learned grasping, motion planning) into a system that handles an unseen ARIAC scenario.

Grading Structure

Assignment Component	Percentage
Individual Assignments	25%
Quizzes	20%
Team Projects	25%
Final Project	30%
Total	**100%**

Final Grade Cutoffs

+	97.00%	+	87.00%	+	77.00%	+	67.00%	
A	94.00%	B	84.00%	C	74.00%	D	64.00%	$F < 60.00\%$
-	90.00%	-	80.00%	-	70.00%	-	60.00%	

Final letter grades are assigned based on the percentage of total assessment points earned. Standards are applied consistently: being close to a cutoff is not the same as making the cut (89.99 is not 90.00).

Academic Integrity

Some assignments will be collected via Turnitin on the course ELMS page to support scholarly writing and to verify the integrity of student work.

The University Code of Academic Integrity is designed to ensure that the principles of academic honesty and integrity are upheld. The University of Maryland does not tolerate academic dishonesty. All acts of academic dishonesty will be dealt with in accordance with the provisions of this code.

Course assistance websites (such as CourseHero) and AI-generated content are not permitted as sources unless the instructor explicitly grants permission. Specifically:

- Using AI tools (including LLM APIs) is required for some assignments and the final project. The expectation is that you understand, design, and debug the code that calls these tools. Submitting AI-generated code that you do not understand is an academic integrity violation.
- For individual assignments, you may use AI tools to clarify concepts or look up syntax, but the submitted code and report must be your own work.
- For quizzes, no external tools or notes are permitted.

On each exam or assignment you must write out and sign the following pledge: *"I pledge on my honor that I have not given or received any unauthorized assistance on this exam/assignment."*

Course Outline

Week	Topic	Deliverables
#1 (TODO: date)	Course Overview and ROS 2 Refresher	Diagnostic quiz (ungraded)
#2 (TODO: date)	System Architectures for AI-Augmented Robotics	Assignment #1 handed out
#3 (TODO: date)	Behavior Trees for Task Execution	Quiz #1
#4 (TODO: date)	Sensing and Perception for Robotic Systems	Quiz #2; Assignment #1 due; Assignment #2 handed out
#5 (TODO: date)	Task-Level Planning with PDDL	
#6 (TODO: date)	Plan Execution Patterns with Behavior Trees	Quiz #3; Assignment #2 due; Assignment #3 handed out
#7 (TODO: date)	LLM-Based Task Planning I: Prompting, Structured Output, Tool Use	
#8 (TODO: date)	SPRING BREAK	
#9 (TODO: date)	LLM-Based Task Planning II: VLM Grounding and Robust Replanning	Quiz #4; Assignment #3 due; Assignment #4 handed out; Final Project handed out
#10 (TODO: date)	Frames, Transforms, and Motion Planning	
#11 (TODO: date)	Learned Grasp Prediction	Quiz #5
#12 (TODO: date)	Evaluation and Measurement Science for AI-Augmented Systems	Assignment #4 due
#13 (TODO: date)	Guest Lecture	
#14 (TODO: date)	Final Project Lab	

Week	Topic	Deliverables
#15 (TODO: date)	Final Project Demo and Presentation	Final Project due

Note: This is a tentative schedule and subject to change as necessary. Monitor the course ELMS page for current deadlines. Week #13 (Guest Lecture) is subject to change based on speaker availability. In the event of a prolonged university closing or extended absence, adjustments to the schedule, deadlines, and assignments will be made based on the duration of the closing and the specific dates missed.

Resources and Accommodations

Accessibility and Disability Services

The University of Maryland is committed to creating and maintaining a welcoming and inclusive educational environment. The Accessibility & Disability Service (ADS) provides reasonable accommodations to qualified individuals. ADS cannot assist retroactively, so it is generally best to request accommodations several weeks before the semester begins or as soon as a disability becomes known. Any student who needs accommodations should contact the instructor as soon as possible.

For assistance, contact ADS at 301-314-7682 or `adsfrontdesk@umd.edu`.

Student Resources and Services

UMD provides a wide range of student support resources. See `https://studentsuccess.umd.edu/` for the full list, including the Writing Center, the Counseling Center, and academic support services.

Notice of Mandatory Reporting

As a faculty member, the instructor is designated as a Responsible University Employee and must report all disclosures of sexual assault, sexual harassment, interpersonal violence, and stalking to UMD's Title IX Coordinator.

For confidential support, contact CARE to Stop Violence at 301-741-3442 or the Counseling Center at 301-314-7651.

You may also contact UMD's Title IX Coordinator at 301-405-1142 or `titleIXcoordinator@umd.edu`.

Basic Needs Security

If you have difficulty affording groceries or accessing sufficient food, or lack a safe and stable place to live, please contact UMD's Division of Student Affairs for information about available resources. Let the instructor know if there is any way the course can help.

Veteran Resources

UMD provides additional support for student veterans through the Office of Veteran Student Life and the Counseling Center. Veterans and active duty military personnel with special circumstances (e.g., deployments, drill requirements, disabilities) are encouraged to communicate these to the instructor in advance.

Participation

Given the hands-on nature of this course, attendance is important to note-taking and your performance. Class discussion is a critical component of learning. Each student is expected to make substantive contributions. Students with a legitimate reason to miss a session should communicate in advance with the instructor, except in case of emergency. Students who miss a session are responsible for learning what they missed.

Course Evaluation

Please submit a course evaluation through Student Feedback on Course Experiences at the end of the semester. All information submitted is confidential.

Copyright Notice

Course materials are copyrighted and may not be reproduced for anything other than personal use without written permission.